Lessons Learned:

Implementing AI Agents for Higher Education Research Compliance

Presented by:

Sabrina Woo and Nathan Layman

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Speakers

Sabrina Woo

University of Missouri
Data Analyst II

Nathan Layman

University of Idaho
Data Scientist

Agenda

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- Background
- AI Overview
- Model Context Protocol
- Agentic AI Evaluation
- Lessons Learned

Background

The Problem

Research administrators must interpret **federal policy** accurately.

Current AI tools:

- Answers without citations
- Potentially outdated or incorrect info

The Need

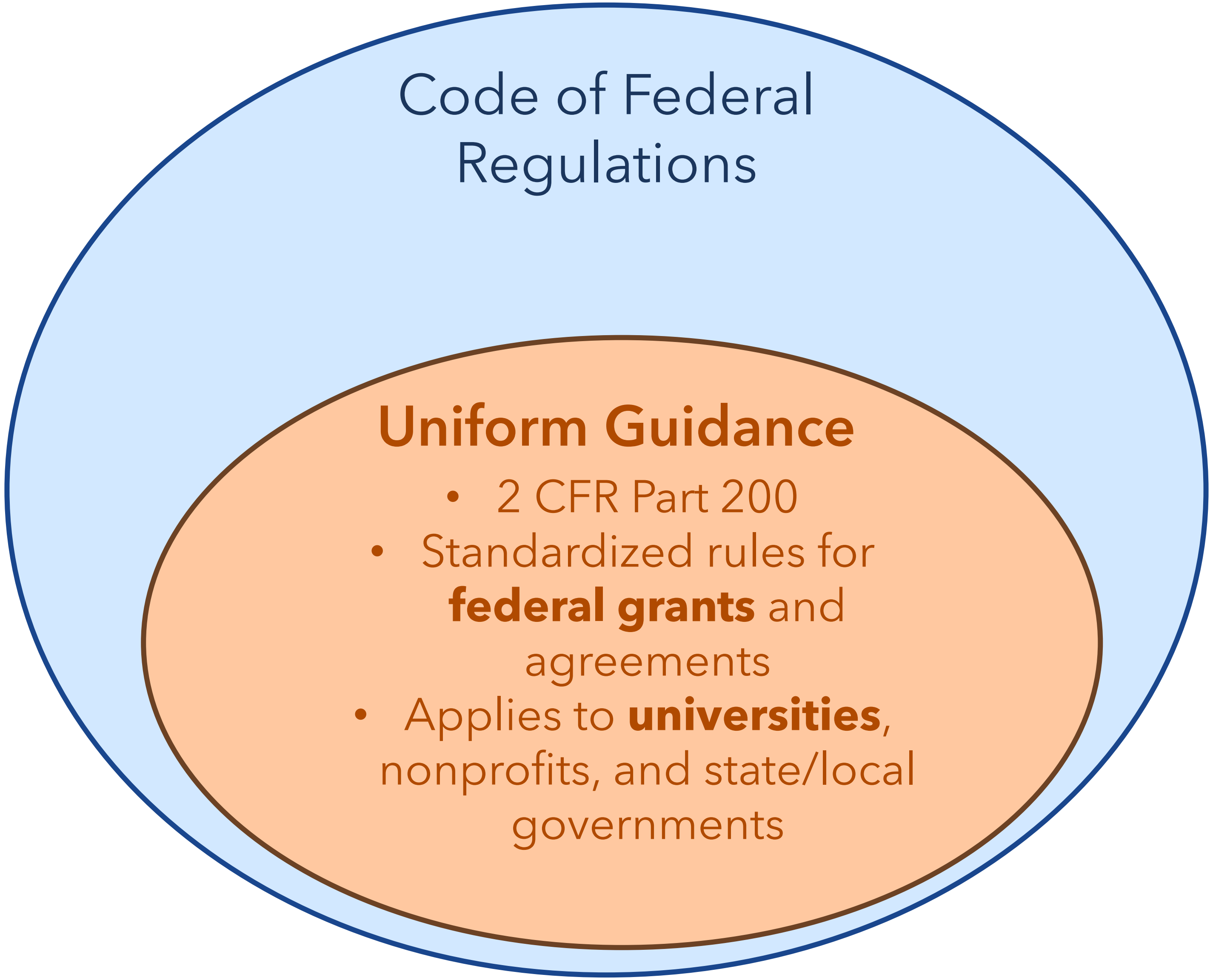
Compliance decisions require:

- Authoritative sources
- Exact documentation

Background

Code of Federal Regulations (CFR)

- Official compilation of **federal rules**
- Organized into **50 subject areas** (Titles)
- Created by federal agencies to implement laws



Background

Our Solution

An MCP-enabled AI that can access the CFR!

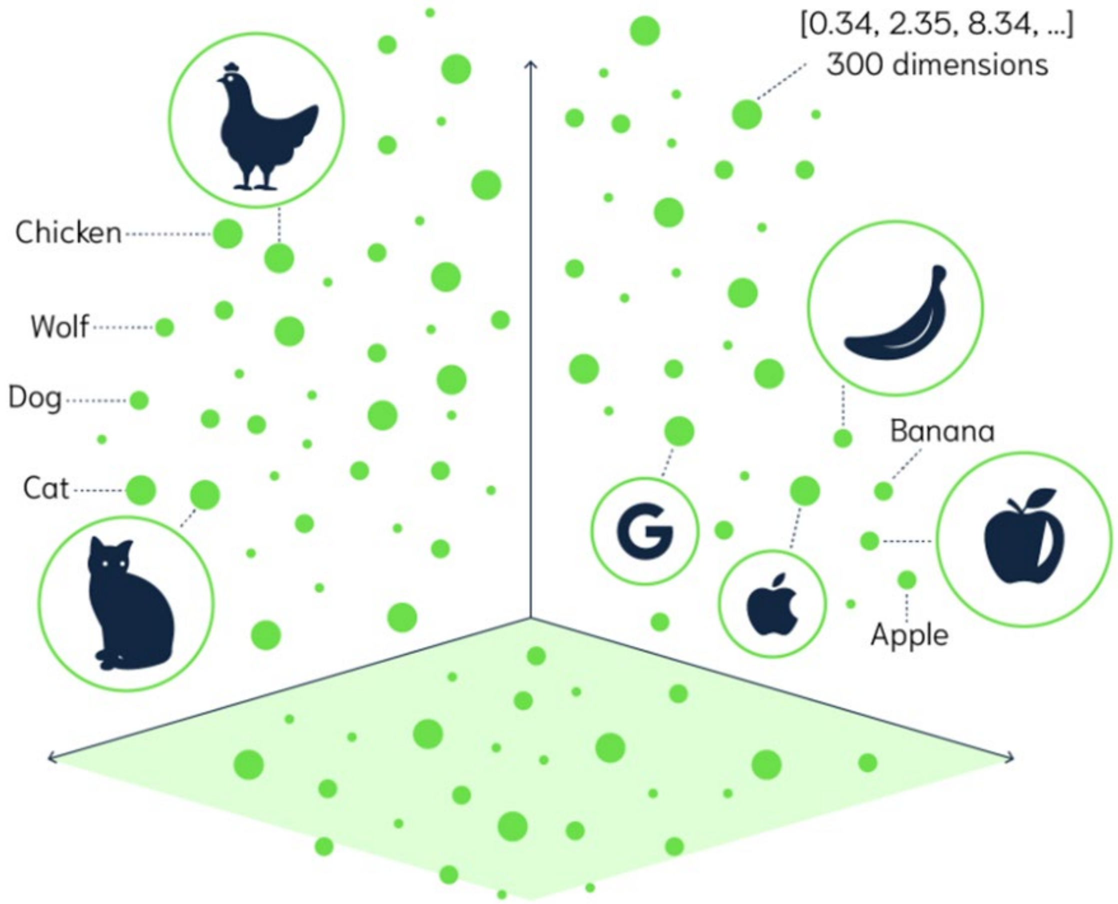
What we built:

Extends an AI's capabilities to support agentic workflows.

What do all these technical terms mean?

AI Overview

AI models are trained to predict and generate sequences of tokens, which can be used to **interpret and transform information.**



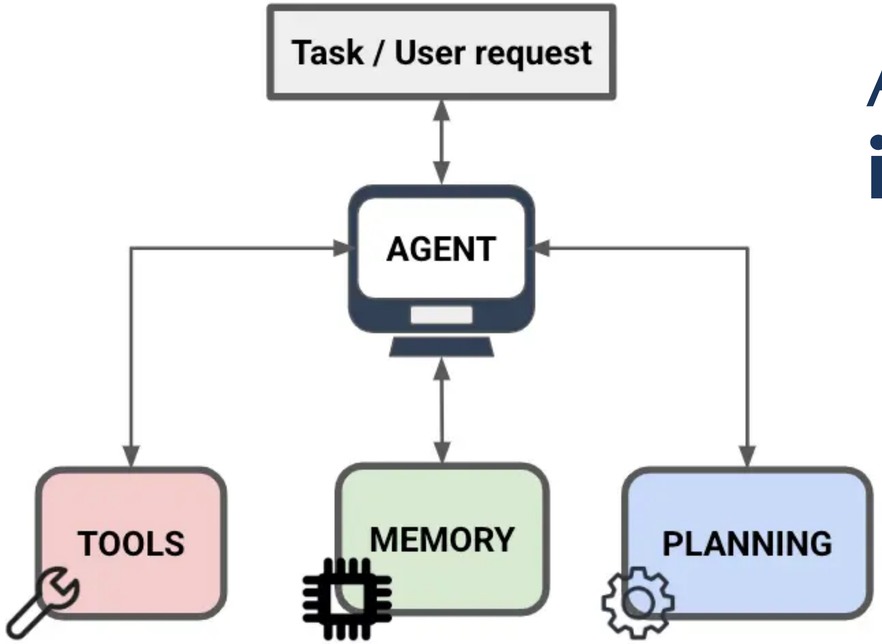
AI Overview

AI models are trained to predict and generate sequences of tokens, which can be used to **interpret and transform information.**

AI Overview

Wordle	LLM
Guess word	Predict next token
Feedback colors	Probability distributions
Iterative solving	Autoregressive generation

AI Agents



An AI **agent** is a system that can **act independently** to achieve a goal.

- Pursues a goal or objective
- Takes independent actions
- Operates over time

Model Context Protocol (MCP)

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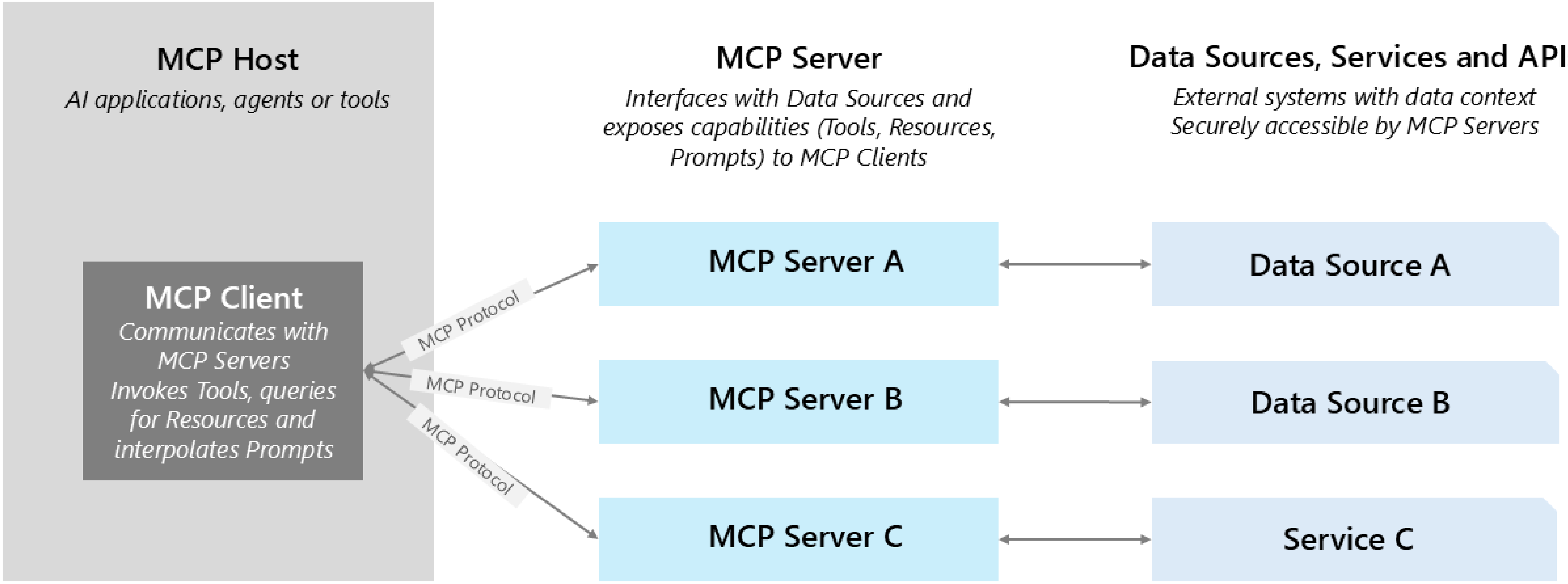
Model Context Protocol acts a **bridge** that connects AI to **external** tools or data

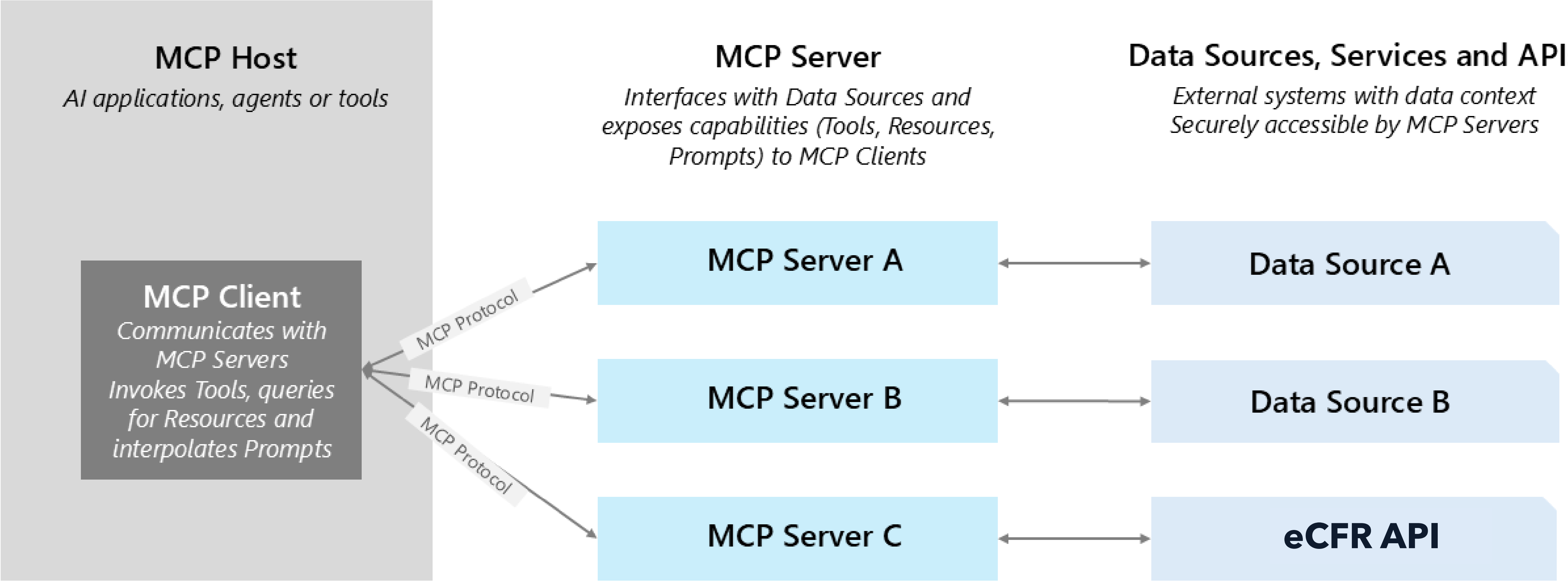
Major Component	Definition
Resources	**Resources** access read-only data.
Prompts	**Prompts** are reusable templates defined by servers that users can select.
Tools	**Tools** are standardized actions declared on the server.

General Framework



Raygun Blog

General Framework



Raygun Blog

Our Framework

Prompts

SKILL.MD

+

Host with MCP Client

Claude Desktop (Sonnet 4.6)

MCP Server

Python, Fast MCP v1

Resources

Regulatory Index

Tools

1. ecfr_search
2. ecfr_get_title_versions
3. ecfr_get_regulation
4. ecfr_compare_regulations
   ecfr_get_title_structure
5. ecfr_list_agencies
6. ecfr_list_titles

Server responses return JSON

Max 1MB

HTTPS Request

Electronic Code of Federal Regulations (eCFR) API

Returns XML Response or JSON


Live Demo

Searched the web >

Under the federal definition established in **2 CFR § 200.68**, the following cost categories must be excluded from the Modified Total Direct Cost (MTDC) base when calculating indirect costs:

1. **Equipment** — equipment and capital expenditures are excluded from MTDC. University of Washington
2. **Patient Care Costs** — charges for patient care are excluded from MTDC. Virginia Department of Cri...
3. **Tuition Remission** — tuition remission is excluded from the MTDC base. University of Washington
4. **Rental Costs** — rental costs are excluded from MTDC. University of Washington
5. **Scholarships and Fellowships** — scholarships and fellowships are excluded from the base. University of Washington
6. **Participant Support Costs** — participant support costs are excluded from MTDC. University of Washington

7. **Subaward/Subcontract Amounts Over $25,000** — the portion of each subaward in excess of $25,000 is excluded Virginia Department of Cri... (only the first $25,000 of each subaward is included in the MTDC base).

Beyond these standard exclusions, other items may only be excluded when necessary to avoid a serious inequity in the distribution of indirect costs, and only with the approval of the cognizant agency for indirect costs. University of California

It's also worth noting that non-federal sponsors that use an MTDC base likely define it differently from the federally-defined MTDC, so it's always critical to understand a sponsor's specific policy. University of California

Used ecfr-mcp integration, viewed a file, loaded tools

Reading the eCFR research admin skill

Loading tools

Result

Ecfr mcp server ecfr get title versions

Result

Ecfr mcp server ecfr search

Result

Ecfr mcp server ecfr search

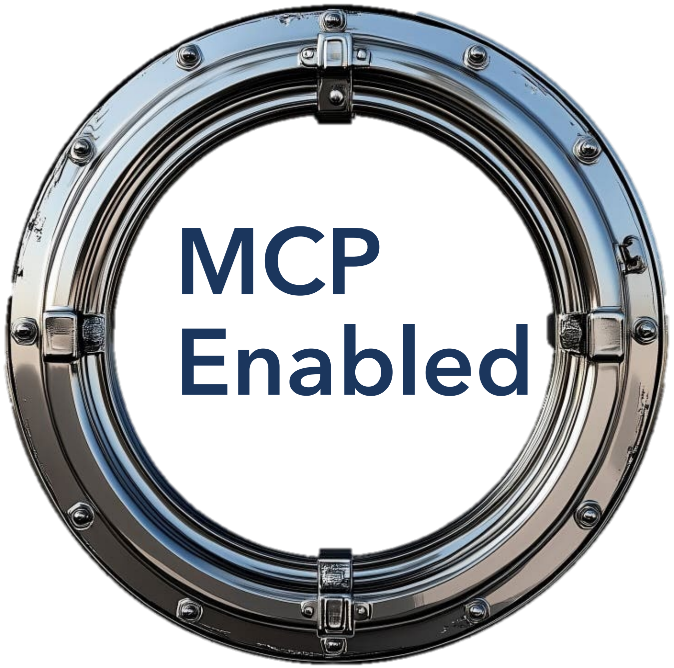
Result

Ecfr mcp server ecfr get regulation

Request


```
{
  "date": "2024-10-01",
  "part": "200",
  "section": "200.1",
  "title": "2",
  "text_only": "true"
}
```


Done



MTDC Exclusions Under the Uniform Guidance

P r **2 CFR § 200.1** (as o October 1, 2024 — the current version of the Uniform Guidance), t e **Modified Total Dir ct Cost (MTDC)** base is defined as:

> *"All direct salaries and wages, applicable fringe benefits, materials and supplies, services, travel, and up to the first **$50,000** of each subaward (regardless of the period of performance of the subawards under the award)."*

Practical implications to keep in mind:

The $50,000 s baward threshold applies *per subaward*, not per project. If you have five subawards ea h totaling $200,000, only the first $50,000 of each ($250,000 total) enters the MTDC base. Also note that "equipment" uses the institution's capitalization threshold (or $10,000, whichever is lower), so items below that threshold may qualify as *supplies* and **would** be included in the base.

If you're working with a specific agency program or negotiating a rate agreement, your cognizant agency (e.g., DHHS, ONR, NSF) may have additional guidance that builds on these regulatory floors. Always verify with your sponsored programs or compliance office for institution-specific application.

AI Evaluation

- **MCP** vs. **Web Search (No MCP)**
- 4 Qs x 10 Repeats (n = 80)
- 6 Metrics
- Human and AI scoring
  - Different AI with More Reasoning
- No Conversation History
- Consistent SKILL.MD

Category	Question
Charging Salary	Can an institution charge salary to a federal award based on an employee's full institutional salary?
Equipment Scenario	I'm working with a PI in Plant Sciences who would like to purchase equipment that was not included in the original approved budget. Could you let me know what prior approval documentation is required and what steps we need to complete before the purchase?
MTDC Exclusion	What expenses must be excluded from the Modified Total Direct Cost base when calculating indirect costs?
Denied Extension	What happens if a project cannot be completed by the end date and a no-cost extension is denied?

Comparison

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MCP Enabled

Pros

- Exact citation texts and traceability
- Specific and consistent responses
- More formal

Cons

- May not provide broader information
- Longer response times
- Higher crash rates

Web Search (No MCP)

Pros

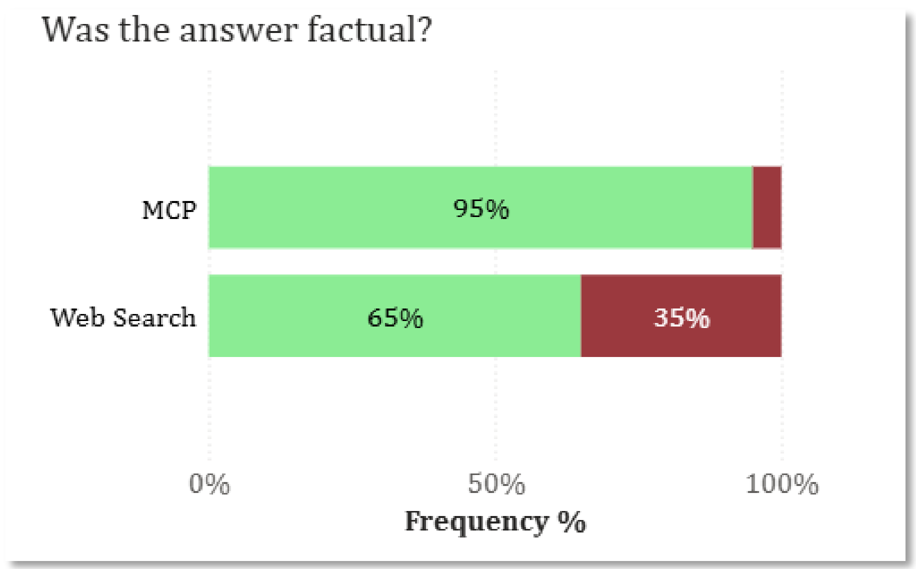
- Incorporates external sponsor guidelines
- Better practicality info
- More casual

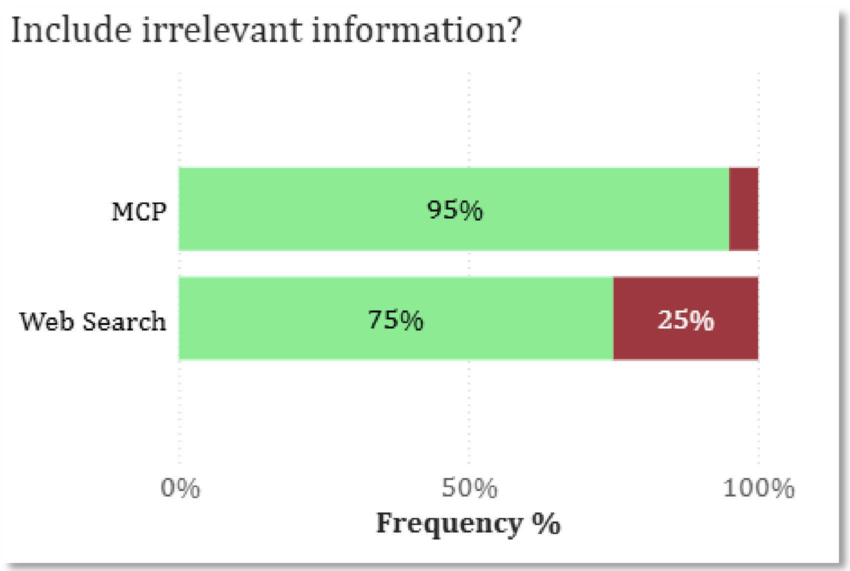
Cons

- May only use training data
- Searches non-governmental websites
- Outdated information

Evaluation Metrics

How does each AI method perform under evaluation?





Chi-squared results:
$\chi 2=7.97$
$p\approx 0.00047$

Chi-squared results:
$\chi 2=3.55$
$p\approx 0.059$

Insights

- Web search performs worse in both metrics.

Evaluation Metrics

How does each AI method perform under evaluation?

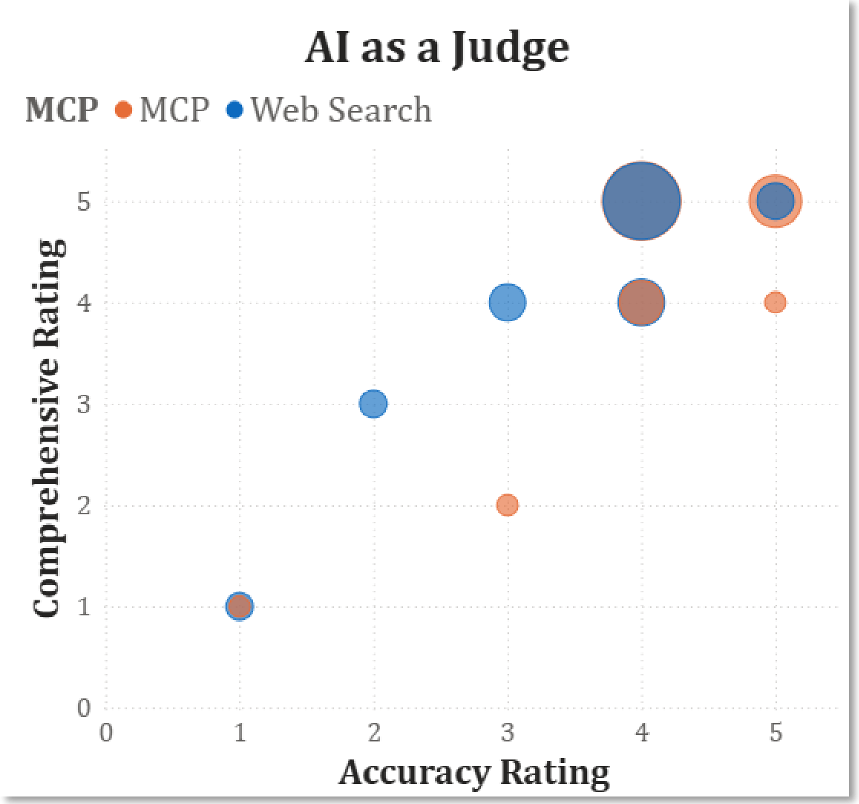
Insights

- Both Human and AI Evaluation gave **better accuracy scores to MCP** generated responses.

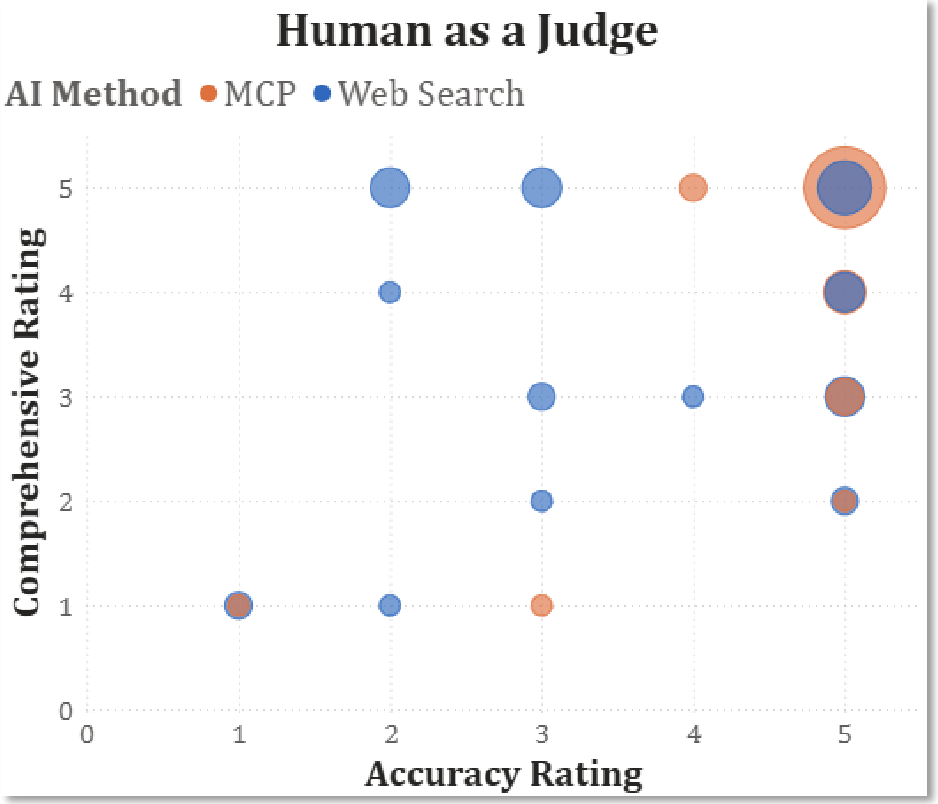
Metric	AVG MCP	AVG Web Search	AVG Diff	t	df	p
Comprehensive by AI	4.65	4.43	0.23	1.10	75.94	0.27
Comprehensive by Human	4.38	3.93	0.45	1.66	75.81	0.10
Accuracy by AI	4.15	3.75	0.40	2.22	73.58	0.03
Accuracy by Human	4.80	3.85	0.95	3.84	58.73	0.0003

Evaluation Metrics

How does a human judge versus an AI judge compare?



Spearman rho = 0.56
p ≈ 9.4e-08



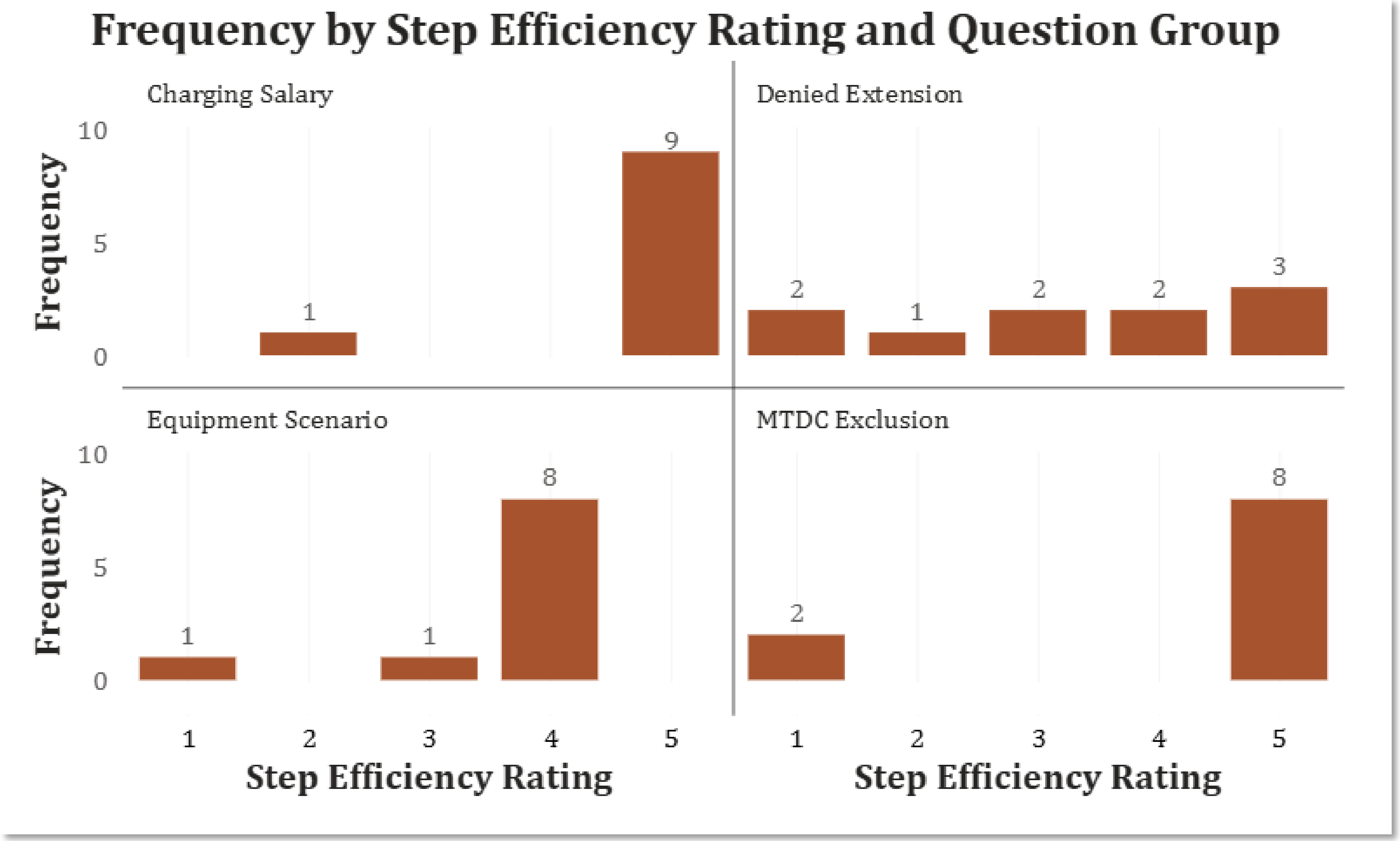
Spearman rho = 0.18
p = 0.115

Insights

- LLMs judge responses have **positive correlation** between comprehensiveness and accuracy.
- Human judge gauges accuracy and comprehensiveness **independently.**

Evaluation Metrics

How efficient is the MCP tool usage by the AI Agent?



Insights

- Question type determines efficiency
- Specific questions perform better
- Questions with what-ifs perform worse


Lessons
Learned

Benefits

Major

- Evidence-based
- Easy to follow citation text and search exact dates
- Extendable and scalable

Other

- Easy to setup and use
- Customizable
- Ability to reuse via cloud hosting
- Public data, no security concerns
- Non-technical user focused

Challenges

Major

- Smaller and older models do not perform well
- Context windows and tool response maximums
- Middleman API response long

Other

- At the whims of AI client providers
- Too many tools leads to agent confusion
- Ensuring that the agent doesn't default to training data
- No high-level overviews

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Future Considerations

Major

- Cloud hosting
- Different AI clients
- Stress testing API
- More human evaluations

Other

- Different context windows
- Not all clients can use MCP
- Agent can execute code and save files
- User prompts
- Other tools (e.g., university SOPs)

Resources

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- https://www.ecfr.gov/developers/documentation/api/v1
- https://huggingface.co/spaces/n8layman/ecfr-mcp-server
- https://ai4ra.uidaho.edu
- https://www.anthropic.com/claude/sonnet
- https://github.com/AI4RA/mcp-ecfr
- https://ai4ra.github.io/mcp-ecfr/

Setup Instructions

bit.ly/48gKQQ2

Thank you

NSF External Evaluation
bit.ly/3NwQpTi

Connect with us!

Sabrina Woo
University of Missouri
Data Analyst II
Email: Sabrina.y.woo@missouri.edu

Nathan Layman
University of Idaho
Data Scientist
Email: nlayman@uidaho.edu